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PowerPoint Outline - VietMach KPI/OKR System

1. Thong Tin Tong Quan

Ten de tai: VietMach KPI/OKR System - He thong quan ly KPI/OKR cho doanh nghiep

Muc dich he thong: So hoa quy trinh quan tri hieu suat tu muc tieu chien luoc, OKR, KPI, check-in tien do, phe duyet, danh gia nhan su, xep hang, thuong du kien, bao cao va AI Insights.

Doi tuong ap dung: Doanh nghiep co nhieu phong ban, nhieu cap quan ly va can theo doi hieu suat nhan su theo ky danh gia.

Thanh vien theo vai tro thuyet trinh:

Vai tro	Noi dung phu trach
PO	Bai toan, khach hang, muc tieu, y tuong, pham vi, backlog
SM	Scrum, Trello/Git, sprint backlog, giam sat tien do, Definition of Done
Luong he thong	Use case, workflow, UI flow, mo hinh phan ra chuc nang
Frontend	Phong cach thiet ke, giao dien, trai nghiem nguoi dung
Backend	Kien truc ky thuat, ERD, database, phan quyen, service, AI
Tester	Test plan, test case, dam bao chat luong, tieu chi nghiem thu

Thoi luong: 20 phut

So slide goi y: 16 slide

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2. Bang Phan Chia Thoi Gian

Phan	Nguoi trinh bay	Thoi luong	Muc tieu
1	PO	3 phut	Gioi thieu bai toan, khach hang, muc tieu va pham vi
2	SM	3 phut	Trinh bay cach nhom quan ly cong viec va kiem soat sprint
3	Luong he thong	4 phut	Giai thich luong nghiep vu, use case, workflow, UI flow
4	Frontend	3 phut	Trinh bay giao dien, phong cach thiet ke va UX
5	Backend	4 phut	Trinh bay kien truc, database, service, bao mat va AI
6	Tester	3 phut	Trinh bay test plan, test case va dam bao chat luong
Tong	**6 nguoi**	**20 phut**	**Hoan chinh bai thuyet trinh**

3. Outline Slide-By-Slide

Slide 1 - Tieu De Va Thanh Vien

Nguoi trinh bay: PO

Thoi luong: 0.5 phut

Noi dung tren slide:

- VietMach KPI/OKR System
- He thong quan ly KPI/OKR cho doanh nghiep
- Nhom thuyet trinh gom 6 vai tro: PO, SM, Luong he thong, Frontend, Backend, Tester
- Muc tieu bai thuyet trinh: giai thich bai toan, giai phap, luong he thong va chat luong san pham

Goi y loi noi:

> Xin chao thay co va cac ban. Nhom em trinh bay de tai VietMach KPI/OKR System, mot he thong web ho tro doanh nghiep quan ly muc tieu, KPI, check-in tien do, phe duyet, danh gia va bao cao hieu suat. Bai trinh bay se duoc chia cho 6 vai tro de the hien day du ca goc nhin san pham, quy trinh, ky thuat, giao dien va kiem thu.

Slide 2 - Bai Toan Va Thu Thap Y Kien Khach Hang

Nguoi trinh bay: PO

Thoi luong: 1 phut

Noi dung tren slide:

- Khach hang muc tieu: doanh nghiep co nhieu phong ban, quan ly nhan su theo KPI/OKR
- Nhu cau thu thap duoc:
  - Can gan muc tieu chien luoc voi KPI hang ngay
  - Can nhan vien check-in tien do minh bach
  - Can quan ly duyet ket qua truoc khi tinh diem
  - Can dashboard, bao cao, xep hang va thuong du kien
  - Can phan quyen ro theo vai tro
- Van de hien tai:
  - KPI thu cong bang Excel hoac file rieng le
  - Kho kiem soat tien do theo thoi gian thuc
  - Du lieu danh gia de sai lech, thieu lich su phe duyet

Goi y loi noi:

> Khi thu thap y kien tu goc nhin doanh nghiep, nhom nhan thay van de lon nhat la viec quan ly KPI/OKR thuong bi tach roi: muc tieu chien luoc mot noi, KPI mot file, check-in mot kenh khac, danh gia lai xu ly thu cong. Dieu nay lam cho nha quan ly kho biet muc tieu nao dang cham, nhan vien nao can ho tro, va ket qua cuoi ky co cong bang hay khong.

Slide 3 - Moi Truong Ap Dung, Quy Mo Va Tam Nhin

Nguoi trinh bay: PO

Thoi luong: 1 phut

Noi dung tren slide:

- Moi truong ap dung:
  - Web noi bo doanh nghiep
  - Bo phan HR, Director, Manager, Employee/Sales
  - Van hanh theo ky danh gia thang/quy/nam
- Quy mo du an:
  - Nhieu module: tai khoan, phong ban, OKR, KPI, check-in, evaluation, report, AI
  - Nhieu vai tro va pham vi du lieu khac nhau
  - Co database, migration, seed data va du lieu demo
- Tam nhin:
  - Bien KPI/OKR thanh quy trinh so hoa lien tuc
  - Ket noi chien luoc cong ty voi hieu suat tung nhan vien
  - Ho tro ra quyet dinh bang bao cao va AI Insights

Goi y loi noi:

> He thong duoc thiet ke cho moi truong doanh nghiep noi bo, noi co cac vai tro nhu Director, Manager, HR va Employee. Quy mo khong chi dung o viec nhap KPI, ma bao gom ca nen tang to chuc, OKR, KPI, check-in, danh gia, bao cao va AI. Tam nhin cua du an la tao mot vong doi quan tri hieu suat ro rang, co du lieu va co kha nang mo rong.

Slide 4 - Chuc Nang Chinh, Phu Va Phi Chuc Nang

Nguoi trinh bay: PO

Thoi luong: 0.5 phut

Noi dung tren slide:

- Chuc nang chinh:
  - Quan ly Mission/Vision, OKR, Key Result
  - Tao, duyet, phan bo KPI
  - Check-in KPI va duyet check-in
  - Tu dong tinh diem, xep hang, thuong du kien
  - Dashboard, bao cao Excel, AI Insights
- Chuc nang phu:
  - Quan ly phong ban, chuc vu, nhan vien, tai khoan, vai tro
  - Quen mat khau OTP email, Google OAuth, avatar, audit log
- Yeu cau phi chuc nang:
  - Bao mat theo role/permission
  - Du lieu tap trung, de truy xuat
  - Giao dien de dung, co dashboard truc quan
  - Co kha nang trien khai tren IIS

Goi y loi noi:

> Ve pham vi, chuc nang loi la quan ly OKR, KPI, check-in, danh gia va bao cao. Cac chuc nang nen tang nhu phong ban, nhan vien, role va permission giup he thong van hanh dung thuc te doanh nghiep. Ngoai ra, nhom cung chu trong cac yeu cau phi chuc nang nhu bao mat, tinh on dinh, kha nang trien khai va trai nghiem nguoi dung.

Slide 5 - Quy Trinh Lam Viec Scrum

Nguoi trinh bay: SM

Thoi luong: 1 phut

Noi dung tren slide:

- Nhom ap dung Scrum de chia nho cong viec
- Vai tro trong nhom:
  - PO xac dinh yeu cau va uu tien backlog
  - SM theo doi sprint, thao go vuong mac
  - Dev Frontend/Backend thuc hien tinh nang
  - Tester kiem tra nghiep vu va loi giao dien

- Vong lap sprint:
  - Planning: chon user story
  - Development: lap trinh va commit Git
  - Review: demo tinh nang
  - Retrospective: rut kinh nghiem va cai tien

Goi y loi noi:

> De quan ly mot du an co nhieu module nhu KPI/OKR, nhom su dung Scrum. Moi sprint tap trung vao mot nhom tinh nang ro rang, vi du sprint cho auth va phan quyen, sprint cho OKR/KPI, sprint cho check-in va evaluation, sprint cho dashboard va AI. Cach lam nay giup nhom vua chia viec ro, vua co the demo san pham sau tung giai doan.

Slide 6 - Quan Ly Backlog, Trello, Git Va Sprint Backlog

Nguoi trinh bay: SM

Thoi luong: 1 phut

Noi dung tren slide:

- Product backlog:
  - User story cho tung module: Auth, HR, OKR, KPI, Check-in, Evaluation, Report, AI
  - Uu tien theo gia tri nghiep vu va phu thuoc ky thuat
- Sprint backlog:
  - Chon cac story co the hoan thanh trong sprint
  - Chia task theo frontend, backend, test
- Trello:
  - Columns: Backlog, To Do, In Progress, Review, Testing, Done
  - Gan nguoi phu trach va deadline
- Git:
  - Commit theo tinh nang
  - Review thay doi truoc khi hop nhat
  - Theo doi lich su sua loi

Goi y loi noi:

> Backlog duoc chia thanh cac user story theo module. Khi vao sprint, nhom tach user story thanh task nho cho frontend, backend va tester. Trello giup theo doi trang thai cong viec, con Git giup quan ly ma nguon, lich su thay doi va phoi hop giua cac thanh vien. Nhung tinh nang quan trong nhu phan quyen, check-in va danh gia duoc uu tien vi anh huong truc tiep den luong nghiep vu.

Slide 7 - Giam Sat Tien Do Va Tieu Chuan Hoan Thanh

Nguoi trinh bay: SM

Thoi luong: 1 phut

Noi dung tren slide:

- Giam sat tien do:
  - Daily update ngan: da lam, dang lam, dang bi chan
  - Kiem tra task tren Trello
  - Kiem tra commit va thay doi tren Git
  - Demo nhanh sau khi hoan thanh module
- Definition of Done:
  - Tinh nang chay duoc tren local
  - Dung nghiep vu va dung phan quyen
  - Co validation du lieu dau vao
  - Giao dien khong vo layout
  - Tester da test case chinh
  - Khong lam hong luong dang co

Goi y loi noi:

> Tieu chuan hoan thanh cua nhom khong chi la code xong. Mot task chi duoc xem la Done khi chay duoc, dung role, dung nghiep vu, co validation va da duoc test. Vi he thong co nhieu vai tro, viec kiem tra phan quyen la bat buoc, neu khong se de gay loi nhu Employee nhin thay du lieu cua Manager hoac tu duyet check-in cua minh.

Slide 8 - Use Case Va Tac Nhan He Thong

Nguoi trinh bay: Luong he thong

Thoi luong: 1 phut

Noi dung tren slide:

- Tac nhan chinh:
  - Admin: quan tri he thong, role, permission, catalog, audit
  - Director: quan ly muc tieu chien luoc, xem dashboard toan cong ty, duyet danh gia cuoi
  - Manager: tao/phan bo KPI, duyet check-in, gui evaluation len Director
  - HR: quan ly nhan su, ky danh gia, bonus rule, bao cao nhan su
  - Employee/Sales: xem KPI duoc giao, check-in tien do, xem ket qua ca nhan
- Use case noi bat:
  - Dang nhap va phan quyen
  - Tao OKR va Key Result
  - Tao, duyet, phan bo KPI
  - Check-in va review check-in
  - Danh gia, xep hang, bao cao
  - AI chat, goi y KPI, phan tich hieu suat

Goi y loi noi:

> He thong co nhieu tac nhan voi quyen khac nhau. Admin quan tri toan bo, Director nhin muc tieu chien luoc, Manager van hanh KPI phong ban, HR phu trach du lieu nhan su va evaluation, con Employee chi thao tac tren du lieu ca nhan. Diem quan trong la cung mot module, moi vai tro co pham vi du lieu va thao tac khac nhau.

Slide 9 - Workflow Nghiep Vu End-To-End

Nguoi trinh bay: Luong he thong

Thoi luong: 1.5 phut

Noi dung tren slide:

```
Mission/Vision
    -> OKR
    -> Key Result
    -> KPI
    -> Phan bo KPI cho phong ban / nhan vien
    -> Nhan vien check-in tien do
    -> Manager/Director/HR/Admin duyet check-in
    -> Cap nhat tien do KPI
    -> Tinh total score theo trong so
    -> Map sang Grading Rank
    -> Tinh Expected Bonus
    -> Tao/Cap nhat Evaluation Result
    -> Gui Director review
    -> Dashboard / Reports / AI Insights
```

- Neu check-in bi tu choi: khong tinh vao diem chinh thuc
- Neu check-in duoc duyet: he thong cap nhat diem, rank va thuong du kien
- Evaluation co vong `PendingDirectorReview` de Director chot ket qua cuoi

Goi y loi noi:

> Luong nghiep vu bat dau tu Mission/Vision, sau do chuyen thanh OKR va Key Result. Tu Key Result, doanh nghiep co the tao KPI theo ky danh gia va phan bo cho phong ban hoac nhan vien. Khi nhan vien check-in, du lieu khong duoc tinh ngay ma di qua buoc review. Sau khi duoc duyet, he thong moi tinh tien do, tong diem, xep hang va thuong du kien, sau do day vao evaluation va bao cao.

Slide 10 - UI Flow Cho Nguoi Dung

Nguoi trinh bay: Luong he thong

Thoi luong: 1 phut

Noi dung tren slide:

- Director:
  - Login -> Dashboard -> OKRs -> EvaluationReports -> ReviewBoard
- Manager:
  - Login -> KPIs -> AllocatePersonnel -> ReviewQueue -> EvaluationResults
- Employee:
  - Login -> Dashboard ca nhan -> KPIs duoc giao -> KPICheckIns/Create
- HR:

    - Login -> Employees -> EvaluationPeriods -> BonusRules -> EvaluationReports
- Luong demo nhanh:
    - Director xem muc tieu va dashboard
    - Employee check-in KPI
    - Manager duyet check-in
    - Director duyet evaluation cuoi

Goi y loi noi:

> Ve UI flow, moi vai tro co duong di khac nhau. Director bat dau tu dashboard va OKR de nam buc tranh toan cong ty. Manager tap trung vao KPI, phan bo va hang cho duyet check-in. Employee chu yeu xem KPI duoc giao va gui check-in. Cach dieu huong nay giup he thong giam nhieu thao tac thua va dam bao nguoi dung chi thay phan viec lien quan.

Slide 11 - Mo Hinh Phan Ra Chuc Nang Va Cau Truc Phan Tang

Nguoi trinh bay: Luong he thong

Thoi luong: 0.5 phut

Noi dung tren slide:

- Mo hinh phan ra chuc nang:
    - Nen tang: Role, Permission, User, Employee, Department, Position
    - Chien luoc: Mission/Vision, OKR, Key Result
    - KPI: KPI, KPI Detail, KPI Assignment
    - Thuc thi: KPI Check-in, Comment, One-on-One
    - Danh gia: EvaluationResult, GradingRank, BonusRule
    - Bao cao va AI: Dashboard, Export Excel, Smart Alerts, Gemini
- Cau truc phan tang:
    - Presentation: Razor Views, CSS, JS
    - Controller: Auth, Dashboard, OKR, KPI, Check-in, Evaluation, AI
    - Service/Helper: Email, Gemini, AIData, Notification, Progress, Encryption
    - Data Access: EF Core, MiniERPDbContext
    - Database: SQL Server

Goi y loi noi:

> Neu phan ra chuc nang, he thong gom cac nhom lon: nen tang to chuc, chien luoc OKR, KPI, check-in, danh gia, bao cao va AI. Ve phan tang ky thuat, nguoi dung thao tac tren Razor Views, request di vao Controller, cac nghiep vu phuc tap duoc tach sang Service va Helper, sau do EF Core lam viec voi SQL Server.

Slide 12 - Phong Cach Thiet Ke Frontend

Nguoi trinh bay: Frontend

Thoi luong: 1 phut

Noi dung tren slide:

- Phong cach: enterprise dashboard, chuyen nghiep, de doc
- Mau chu dao: xanh duong doanh nghiep, nen sang, sidebar xanh dam
- Thanh phan giao dien:
  - Sidebar co module theo quyen
  - Header va dashboard cards
  - Bang du lieu, form nhap lieu, modal, alert
  - Bieu do thong ke KPI/OKR/check-in
- Thu vien giao dien:
  - Razor Views
  - Bootstrap
  - jQuery
  - Select2
  - Chart.js
  - SweetAlert2

Goi y loi noi:

> Ve frontend, nhom chon phong cach enterprise vi he thong phuc vu quan ly noi bo. Mau xanh duong tao cam giac chuyen nghiep va phu hop voi dashboard quan tri. Sidebar la thanh dieu huong chinh, nhung noi dung hien thi theo role/permission, vi vay moi nguoi dung chi thay cac module duoc cap quyen.

Slide 13 - Giao Dien He Thong

Nguoi trinh bay: Frontend

Thoi luong: 1.25 phut

Noi dung tren slide:

- Dashboard:
  - Tong quan KPI/OKR/check-in theo ky danh gia
  - Bieu do trang thai, xu huong va top hieu suat
- OKR:
  - Danh sach OKR, Key Result, tien do
  - Gan Mission/Vision, phong ban hoac nhan vien
- KPI:
  - Tao KPI, duyet/tuchoi KPI, phan bo KPI theo trong so
  - Lien ket KPI voi OKR/Key Result
- Check-in:
  - Employee nhap gia tri thuc dat va ghi chu
  - ReviewQueue cho Manager/Director/HR/Admin
- Evaluation Reports:
  - Tong hop ket qua, rank, expected bonus
  - Export Excel bang EPPlus

Goi y loi noi:

> Cac man hinh chinh duoc thiet ke theo tung luong nghiep vu. Dashboard giup xem tong quan, OKR the hien muc tieu chien luoc, KPI la noi quan ly chi tieu chi tiet, check-in la noi nhan vien cap nhat tien do, va Evaluation Reports la noi tong hop ket qua cuoi ky. Giao dien uu tien bang, form va bieu do vi day la dang du lieu quan ly can doc nhanh va so sanh nhanh.

Slide 14 - Trai Nghiem Nguoi Dung Va Kiem Soat Loi

Nguoi trinh bay: Frontend

Thoi luong: 0.75 phut

Noi dung tren slide:

- Dieu huong theo vai tro:
  - Menu an/hien theo permission
  - Trang truy cap trai phep chuyen den AccessDenied
- Form va validation:
  - Bat buoc truong quan trong
  - Kiem tra du lieu KPI, ky danh gia, gia tri check-in
- Phan hoi nguoi dung:
  - Alert khi tao/sua/xoa/duyet
  - Thong bao notification
  - Modal lich su AI va widget AI chat
- Muc tieu UX:
  - Giam thao tac thua
  - Lam ro trang thai: Pending, Approved, Rejected, Draft
  - Dam bao nguoi dung hieu minh dang o buoc nao cua workflow

Goi y loi noi:

> Do he thong co nhieu trang thai nhu Pending, Approved, Rejected va Draft, frontend phai lam ro trang thai cho nguoi dung. Ngoai ra, menu theo quyen giup tranh viec nguoi dung thao tac nham module. Cac form duoc validation de giam loi du lieu, va cac alert/thong bao giup nguoi dung biet thao tac da thanh cong hay can sua lai.

Slide 15 - Kien Truc Ky Thuat, ERD Va Backend

Nguoi trinh bay: Backend

Thoi luong: 2.75 phut

Noi dung tren slide:

- Tech stack:
  - ASP.NET Core MVC, .NET 10
  - Entity Framework Core 10
  - SQL Server

    - Cookie Authentication, Google OAuth
    - EPPlus export Excel
    - Gemini API cho AI
- Kien truc request:
    - Browser -> Middleware -> Controller -> Service/Helper -> MiniERPDbContext -> SQL Server
    - Razor Views render ket qua ve browser
- Cac nhom bang ERD:
    - Foundation: Roles, Permissions, Role_Permissions, SystemUsers, Employees, Departments, Positions
    - OKR: MissionVisions, OKRs, OKRKeyResults, OKR_Mission_Mappings, OKR_Department_Allocations, OKR_Employee_Allocations
    - KPI: EvaluationPeriods, KPITypes, KPIProperties, KPIs, KPIDetails, KPI_Department_Assignments, KPI_Employee_Assignments
    - Check-in: KPICheckIns, CheckInDetails, CheckInStatuses, CheckInHistoryLogs, GoalComments
    - Evaluation/HR: EvaluationResults, GradingRanks, BonusRules, RealtimeExpectedBonuses, HRExportReports
    - System/AI: SystemAlerts, AuditLogs, AIGenerationHistories
- Phan quyen:
    - `[Authorize]` yeu cau dang nhap
    - `[HasPermission("PERMISSION_CODE")]` chan thao tac theo permission
    - Claim role va PermissionClaimsTransformation bo sung quyen
- Service:
    - GeminiService: goi Gemini API
    - AIDataService: gom ngu canh KPI/OKR theo quyen
    - AIAlertService: tao smart alerts
    - NotificationService: thong bao
    - OKRProgressService/ProgressHelper: tinh tien do
    - EmailService: OTP va email
- Bao mat va cau hinh:
    - Cookie HttpOnly, SameSite, SecurePolicy theo environment
    - Antiforgery token cho form
    - Data Protection keys de cookie song qua recycle IIS
    - Environment config qua `.env` va appsettings

Goi y loi noi:

> Ve backend, he thong dung ASP.NET Core MVC nen request di tu browser qua middleware, vao controller, sau do controller goi service hoac DbContext. EF Core lam viec voi SQL Server, con Razor Views render giao dien. Database duoc chia theo nhom nghiep vu: nen tang to chuc, OKR, KPI, check-in, evaluation, bao cao va AI. Diem quan trong la phan quyen khong chi o giao dien ma nam ca o controller thong qua Authorize va HasPermission. Vi vay ngay ca khi nguoi dung goi truc tiep URL, he thong van kiem tra quyen truoc khi xu ly.

Slide 16 - Test Plan, Test Case Va Ket Luan

Nguoi trinh bay: Tester

Thoi luong: 3 phut

Noi dung tren slide:

- Muc tieu kiem thu:
  - Dam bao dung luong nghiep vu KPI/OKR
  - Dam bao dung phan quyen tung vai tro
  - Dam bao du lieu tinh diem, rank va bonus chinh xac
  - Dam bao giao dien de dung va khong loi luong chinh
- Test plan:
  - Functional test theo module
  - Role/permission test
  - Workflow test end-to-end
  - UI/UX test
  - Report/export test
  - AI fallback test khi thieu API key hoac Gemini loi
- Test case mau:
  - Employee dang nhap va chi thay KPI duoc giao
  - Employee check-in KPI, trang thai la Pending
  - Manager duyet check-in cua nhan vien phong ban minh
  - Manager khong duoc duyet check-in cua chinh minh
  - Check-in Approved cap nhat score, rank, expected bonus
  - Director duyet EvaluationResult o trang thai PendingDirectorReview
  - Role khong co permission bi chuyen den AccessDenied
  - Export Excel tao duoc file bao cao dung ky danh gia
- Ket luan:

- He thong quan ly tron vong doi KPI/OKR
- Du lieu tap trung, minh bach, co phan quyen
- Ho tro dashboard, bao cao va AI Insights
- Co the mo rong cho nhieu doanh nghiep va phong ban

Goi y loi noi:

> Ve kiem thu, nhom tap trung vao ba nhom rui ro lon: sai nghiep vu, sai phan quyen va sai tinh diem. Vi he thong quan ly KPI/OKR co anh huong den danh gia nhan su, mot loi nho nhu tinh sai trong so hoac cho sai vai tro duyet check-in se anh huong den ket qua cuoi ky. Vi vay test case duoc viet theo workflow thuc te: Employee check-in, Manager duyet, he thong tinh diem, Director duyet evaluation va bao cao cap nhat. Ket luan lai, VietMach KPI/OKR System khong chi la noi nhap KPI, ma la he thong quan ly tron vong doi tu chien luoc, thuc thi, phe duyet, danh gia den bao cao va AI.

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4. Noi Dung Thuyet Trinh Chi Tiet Theo Vai Tro

4.1. PO - Gioi Thieu Bai Toan, Khach Hang, Muc Tieu Va Pham Vi

Thoi luong: 3 phut

Noi dung can trinh bay:

> Phan dau tien, em xin trinh bay goc nhin san pham. Du an cua nhom la VietMach KPI/OKR System, mot he thong web giup doanh nghiep quan ly KPI va OKR theo mot quy trinh lien tuc.
>
> Qua viec phan tich nhu cau nguoi dung, nhom nhan thay nhieu doanh nghiep van quan ly KPI bang Excel, file rieng le hoac trao doi thu cong. Cach lam nay co ba van de: thu nhat, muc tieu chien luoc va KPI hang ngay khong duoc ket noi ro; thu hai, nhan vien check-in tien do nhung quan ly kho theo doi trang thai; thu ba, ket qua danh gia cuoi ky de thieu minh bach vi khong co lich su review day du.

> Vi vay, nhom de xuat mot he thong tap trung, trong do doanh nghiep co the tao Mission/Vision, tao OKR, chia nho thanh Key Result, sau do gan KPI cho phong ban hoac nhan vien. Nhan vien se check-in tien do, quan ly duyet ket qua, he thong tinh diem, xep hang, thuong du kien va tao bao cao.
>
> Doi tuong su dung gom Admin, Director, Manager, HR va Employee/Sales. Moi vai tro co pham vi quyen rieng. Vi du Employee chi xem KPI cua minh, Manager quan ly KPI phong ban, Director nhin toan canh va duyet ket qua cuoi, HR quan ly nhan su va bao cao.
>
> Pham vi chinh cua du an gom quan ly OKR, KPI, check-in, evaluation, dashboard, bao cao Excel va AI Insights. Ngoai ra co cac chuc nang nen tang nhu tai khoan, role, permission, nhan vien, phong ban, chuc vu, audit log, quen mat khau qua email va Google OAuth.

Cau chuyen tiep sang SM:

> Sau khi xac dinh bai toan va pham vi san pham, nhom can mot cach quan ly cong viec de dam bao cac module duoc phat trien dung tien do. Phan tiep theo se do vai tro Scrum Master trinh bay.

4.2. SM - Quy Trinh Lam Viec, Backlog Va Definition Of Done

Thoi luong: 3 phut

Noi dung can trinh bay:

> Voi vai tro Scrum Master, em trinh bay cach nhom to chuc cong viec. Vi he thong co nhieu module va moi module lai lien quan den nhau, nhom ap dung Scrum de chia nho thanh cac sprint.
>
> Product backlog duoc xay dung tu cac user story. Vi du: voi Employee, can story "nhan vien co the xem KPI duoc giao va gui check-in"; voi Manager, can story "quan ly co the duyet check-in cua nhan vien trong phong ban"; voi Director, can story "giam doc co the xem dashboard toan cong ty va duyet ket qua evaluation".
>
> Khi vao sprint backlog, nhom chia story thanh task cu the cho frontend, backend va tester. Backend xu ly controller, service, database va phan quyen. Frontend xu

> ly giao dien Razor, bang, form, dashboard. Tester viet test case theo luong nghiep vu va role.
>
> Nhom theo doi tien do bang Trello voi cac cot Backlog, To Do, In Progress, Review, Testing va Done. Git duoc dung de quan ly ma nguon, commit theo tinh nang va kiem tra thay doi truoc khi hop nhat.
>
> Definition of Done cua nhom gom: tinh nang chay duoc, dung yeu cau, dung phan quyen, co validation, giao dien khong vo layout, tester da kiem tra test case chinh, va khong lam hong cac luong da co. Dac biet, voi he thong nay, phan quyen la tieu chi bat buoc vi moi vai tro co pham vi du lieu khac nhau.

Cau chuyen tiep sang Luong he thong:

> Sau khi nhom da quan ly duoc quy trinh phat trien, phan tiep theo se di vao luong he thong de thay cac module lien ket voi nhau nhu the nao.

4.3. Luong He Thong - Use Case, Workflow, UI Flow Va Phan Ra Chuc Nang

Thoi luong: 4 phut

Noi dung can trinh bay:

> Em se trinh bay luong he thong. He thong co cac tac nhan chinh la Admin, Director, Manager, HR va Employee/Sales. Admin quan tri he thong, role, permission va danh muc. Director quan ly muc tieu chien luoc, xem dashboard toan cong ty va duyet evaluation cuoi. Manager tao va phan bo KPI, duyet check-in va gui evaluation len Director. HR quan ly nhan su, ky danh gia va bao cao. Employee/Sales xem KPI duoc giao va check-in tien do.
>
> Luong nghiep vu end-to-end bat dau tu Mission/Vision. Tu muc tieu chien luoc nay, he thong tao OKR va Key Result. Sau do KPI duoc tao theo ky danh gia va co the lien ket voi OKR hoac Key Result. KPI sau khi duoc duyet se duoc phan bo cho phong ban hoac nhan vien kem trong so.
>
> Khi nhan vien check-in, ban ghi mac dinh o trang thai Pending. Manager, Director, HR hoac Admin se review. Neu bi Rejected, check-in khong duoc tinh vao diem chinh thuc. Neu Approved, he thong cap nhat tien do KPI, tinh total score theo

> trong so, map sang GradingRank, tinh Expected Bonus va tao hoac cap nhat EvaluationResult.
>
> EvaluationResult sau do co the duoc Manager gui len Director review. Director se duyet hoac tu choi. Khi duoc duyet, ket qua cuoi ky co the hien thi tren dashboard va bao cao.
>
> Ve UI flow, Director thuong vao Dashboard, OKRs va EvaluationReports. Manager vao KPIs, AllocatePersonnel, ReviewQueue va EvaluationResults. Employee vao Dashboard ca nhan, KPIs va KPICheckIns/Create. HR vao Employees, EvaluationPeriods, BonusRules va EvaluationReports.
>
> Neu phan ra chuc nang, he thong gom cac nhom: nen tang to chuc, chien luoc OKR, KPI, thuc thi check-in, danh gia nhan su, bao cao va AI. Neu phan tang ky thuat, he thong gom Presentation layer, Controller layer, Service/Helper layer, Data Access layer, Database layer va External Services.

Cau chuyen tiep sang Frontend:

> Khi cac luong nghiep vu da ro, frontend se la phan giup nguoi dung thao tac voi cac luong do mot cach truc quan va dung vai tro.

4.4. Frontend - Thiet Ke Giao Dien Va Trai Nghiem Nguoi Dung

Thoi luong: 3 phut

Noi dung can trinh bay:

> Ve frontend, nhom xay dung giao dien theo phong cach enterprise dashboard. Ly do la he thong phuc vu quan ly noi bo, nen giao dien can ro rang, de doc va ho tro xem so lieu nhanh.
>
> Mau sac chu dao la xanh duong chuyen nghiep, nen sang, sidebar xanh dam. Sidebar giup dieu huong giua cac module nhu Dashboard, OKR, KPI, Check-in, Evaluation, Employees va Reports. Diem quan trong la sidebar khong hien thi giong nhau cho moi nguoi, ma an/hien theo permission.

> Cac man hinh chinh gom Dashboard, OKRs, KPIs, KPICheckIns, ReviewQueue, EvaluationResults va EvaluationReports. Dashboard dung card va bieu do de hien thi KPI/OKR/check-in theo ky danh gia. OKR va KPI dung bang du lieu va form de tao, sua, duyet, phan bo. Check-in co form nhap gia tri thuc dat va ghi chu. ReviewQueue giup quan ly duyet cac check-in dang Pending.
>
> Ve trai nghiem nguoi dung, nhom chu trong cac trang thai nghiep vu nhu Pending, Approved, Rejected, Draft va PendingDirectorReview. Cac trang thai nay can hien thi ro de nguoi dung biet du lieu dang o buoc nao. He thong cung co alert, notification, validation form va trang AccessDenied khi nguoi dung khong co quyen.
>
> Frontend su dung Razor Views ket hop Bootstrap, jQuery, Select2, Chart.js va SweetAlert2. Cach ket hop nay giup giao dien du on dinh, de phat trien nhanh va phu hop voi mo hinh ASP.NET Core MVC.

Cau chuyen tiep sang Backend:

> Phia sau cac man hinh nay la backend xu ly nghiep vu, database, phan quyen va tich hop AI. Phan tiep theo se trinh bay kien truc ky thuat cua he thong.

4.5. Backend - Kien Truc Ky Thuat, ERD, Phan Quyen Va AI

Thoi luong: 4 phut

Noi dung can trinh bay:

> Ve backend, he thong duoc xay dung bang ASP.NET Core MVC tren .NET 10, dung Entity Framework Core 10 de lam viec voi SQL Server. He thong co cookie authentication, Google OAuth, export Excel bang EPPlus, va tich hop Gemini API cho cac tinh nang AI.
>
> Luong request co the hieu nhu sau: nguoi dung thao tac tren browser, request di qua middleware nhu HTTPS, routing, authentication, authorization va antiforgery. Sau do request vao Controller. Controller se goi service, helper hoac MiniERPDbContext. EF Core truy van SQL Server va Razor Views render ket qua tra ve giao dien.

> Ve ERD, database duoc chia thanh nhieu nhom bang. Nhom foundation gom Roles, Permissions, Role_Permissions, SystemUsers, Employees, Departments va Positions. Nhom OKR gom MissionVisions, OKRs, OKRKeyResults va cac bang allocation. Nhom KPI gom EvaluationPeriods, KPITypes, KPIProperties, KPIs, KPIDetails va KPI assignments. Nhom check-in gom KPICheckIns, CheckInDetails, CheckInStatuses, CheckInHistoryLogs va GoalComments. Nhom evaluation gom EvaluationResults, GradingRanks, BonusRules va RealtimeExpectedBonuses. Ngoai ra co AuditLogs, SystemAlerts va AIGenerationHistories.
>
> Phan quyen la diem quan trong cua backend. He thong dung `[Authorize]` de yeu cau dang nhap va `[HasPermission("PERMISSION_CODE")]` de chan thao tac theo permission. Permission duoc gan qua Role_Permission. Vi vay, khong chi giao dien an menu, ma backend cung chan truc tiep neu nguoi dung khong co quyen.
>
> Cac service chinh gom GeminiService de goi AI, AIDataService de gom ngu canh KPI/OKR theo quyen, AIAlertService de tao smart alert, NotificationService de thong bao, EmailService de gui OTP/email, va OKRProgressService de tinh tien do OKR. AI chi la lop ho tro phan tich va goi y, khong thay the logic nghiep vu loi.
>
> Ve bao mat va cau hinh, he thong dung cookie HttpOnly, SameSite, SecurePolicy theo environment, antiforgery token cho form, Data Protection keys de cookie khong bi mat khi IIS recycle, va cau hinh qua `.env`, appsettings va environment variables.

Cau chuyen tiep sang Tester:

> Khi he thong co nhieu luong nghiep vu, nhieu role va nhieu tinh toan, kiem thu la phan rat quan trong de dam bao san pham dang tin cay.

4.6. Tester - Test Plan, Test Case Va Dam Bao Chat Luong

Thoi luong: 3 phut

Noi dung can trinh bay:

> Ve kiem thu, muc tieu cua nhom la dam bao he thong dung nghiep vu KPI/OKR, dung phan quyen, dung tinh toan va on dinh ve giao dien.
>
> Test plan duoc chia thanh nhieu nhom. Functional test kiem tra tung module nhu Auth, Employees, OKRs, KPIs, KPICheckIns, EvaluationResults va Reports. Role/permission test kiem tra moi vai tro chi duoc thay va thao tac dung pham vi. Workflow test kiem tra luong end-to-end tu Mission/Vision den OKR, KPI, check-in, review, evaluation va report. UI/UX test kiem tra form, bang, sidebar, alert va responsive. Report test kiem tra export Excel. AI test kiem tra chat, goi y va fallback khi thieu API key hoac Gemini loi.
>
> Mot so test case quan trong gom: Employee dang nhap chi thay KPI duoc giao; Employee tao check-in thi trang thai la Pending; Manager duyet check-in cua nhan vien trong phong ban; Manager khong duoc duyet check-in cua chinh minh; Check-in Approved phai cap nhat score, rank va expected bonus; Director duyet EvaluationResult o trang thai PendingDirectorReview; role khong co permission bi chuyen den AccessDenied; export Excel tao duoc bao cao dung ky danh gia.
>
> Definition of Done cua tester la test case chinh pass, khong co loi nghiem trong tren workflow, khong co loi phan quyen, va cac loi giao dien chinh da duoc sua. Vi he thong lien quan den danh gia nhan su, tester uu tien cac loi co the lam sai ket qua cuoi ky hoac lo du lieu khong dung pham vi.
>
> Ket luan, VietMach KPI/OKR System quan ly tron vong doi tu chien luoc den thuc thi, phe duyet, danh gia va bao cao. He thong giup du lieu KPI/OKR tap trung, minh bach, co phan quyen va co AI ho tro phan tich.

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5. Goi Y So Do De Dua Vao PowerPoint

5.1. So Do Workflow

```
Mission/Vision
  -> OKR
  -> Key Result
  -> KPI
  -> KPI Assignment
  -> KPI Check-in
  -> Review
  -> Score/Rank/Bonus
```

```
  -> Evaluation Result
  -> Director Review
  -> Dashboard/Report/AI
```

5.2. So Do Kien Truc

```
Browser
  -> ASP.NET Core Middleware
  -> Controllers
  -> Services / Helpers
  -> MiniERPDbContext
  -> SQL Server

External:
  -> Gemini API
  -> SMTP Email
```

5.3. So Do ERD Rut Gon

```
Role --< SystemUser -- Employee >-- Department
Role --< Role_Permission >-- Permission

MissionVision --< OKR --< OKRKeyResult
OKR --< KPI --< KPIDetail
KPI --< KPI_Employee_Assignment >-- Employee
KPI --< KPICheckIn >-- Employee
KPICheckIn --< CheckInDetail

Employee --< EvaluationResult >-- EvaluationPeriod
EvaluationResult -- GradingRank
GradingRank --< BonusRule
Employee --< RealtimeExpectedBonus
```

5.4. So Do Phan Tang

```
Presentation Layer:
  Razor Views, Bootstrap, jQuery, CSS, Chart.js

Controller Layer:
  Auth, Dashboard, OKRs, KPIs, KPICheckIns, EvaluationResults, AI

Service/Helper Layer:
```

```
    GeminiService, AIDataService, EmailService, NotificationService, ProgressHelper

  Data Access Layer:
    Entity Framework Core, MiniERPDbContext

  Database Layer:
    SQL Server

  External Services:
    Gemini API, SMTP Email, Google OAuth
```

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6. Checklist Truoc Khi Thuyet Trinh

- PO nam ro bai toan, khach hang, muc tieu va pham vi.
- SM nam ro cach noi ve Scrum, Trello, Git va Definition of Done.
- Luong he thong tap noi workflow Mission/Vision -> OKR -> KPI -> Check-in -> Evaluation -> Report.
- Frontend chuan bi anh chup man hinh Dashboard, OKR, KPI, Check-in, ReviewQueue, EvaluationReports neu co.
- Backend chuan bi so do kien truc va ERD rut gon.
- Tester chuan bi 5-8 test case quan trong nhat.
- Ca nhom tap chuyen tiep giua cac vai tro de bai noi khong bi dut mach.
- Tong thoi gian tap thu nen nam trong khoang 18-20 phut.